

Department of Computing, Mathematical & Statistical Sciences

Bachelor of Science in Data Science Honours(33BHDS)

Big Data Analytics and Data Mining (S4822DB)

Executive Report

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RETAILMART CUSTOMER INTELLIGENCE PLATFORM

Executive Report

To: RetailMart Executive Leadership
From: Data Science Team
Date: October 2026
Subject: Customer Behaviour Insights, Churn Prediction, and Retention Strategy

1. Executive Summary

RetailMart faces declining customer loyalty across its 500+ stores and e-commerce platform. Over a 3-day sprint, our team engineered a complete Customer Intelligence Platform using PySpark and Tableau, processing 5 years of data (1,000,000 transactions and 250,000 feedback records). We developed a scalable data architecture, identified key customer behaviours, and trained machine learning models to predict churn.

Our findings recommend deploying a Gradient-Boosted Trees (GBT) model to accurately flag at-risk customers and a targeted marketing strategy focusing on protecting a VIP segment and winning back two high-churn clusters.

2. Data Quality and Architecture

Business Question: What is the quality of our data, and how should we store and process it at scale?

- Data Quality: The raw transaction dataset contained significant quality issues, including missing values (e.g., 4% missing gender, 3% missing age), duplicate records (10,000 duplicate IDs), invalid dates (4,946), and out-of-range/negative values. After applying rigorous cleaning rules and median imputations, we retained 973,266 high-quality transaction records (97.3% retention) and 250,000 clean feedback records with zero missing values.
- Architecture & Storage: For scalable processing, we migrated the cleaned data into PySpark, utilizing Parquet and ORC formats. Parquet reduced the storage footprint by 59.6% compared to CSV (57.55 MB vs. 142.42 MB). Furthermore, partitioning the data by year and month improved query performance by 87% (reducing query time from 3.89s to 0.50s for a January 2026 lookup), ensuring the platform can scale efficiently.

3. Customer Behaviour Visualisation

Business Question: What insights can we visualise from customer behaviour?

Using Tableau, we identified several critical trends:
- Revenue Drivers: Electronics is our dominant category, generating $76.7 million (nearly 36% of total revenue), driven by a high average transaction value of $629. Conversely, while Groceries generate high volume, the average transaction value is only $31.
- Churn Profile: The baseline churn rate is 22.3%. Customers who churned averaged shorter loyalty durations (1.59 years vs. 2.68) and fewer visits (8.05 vs. 10.41) compared to retained customers.
- Customer Sentiment: Customer feedback averages exactly 3.0 out of 5. While ratings are generally balanced, the highest volume of neutral (3-star) ratings (30%) suggests a "stagnant" customer experience where shoppers are neither delighted nor disappointed.

4. Churn Prediction

Business Question: Can we predict which customers will stop shopping with us?
Yes. We benchmarked four PySpark MLlib classification models on a leak-free 80/20 customer split, using a class weight of 3.48 to handle the 22.3% minority churn class.
- The Winner: The Gradient-Boosted Trees (GBT) model performed best, achieving a ROC-AUC of 0.713 and an accuracy of 70.2%. Crucially, GBT trains in 113 seconds, less than half the time of a Random Forest (257 seconds), making it highly efficient for batch scoring.
- The Failure Mode: A standalone Decision Tree achieved 70% accuracy but collapsed to a random-guess ROC-AUC of 0.50. Its probabilities cannot rank customers properly, disqualifying it for marketing prioritisation.
- Key Drivers: Feature importance analysis revealed that Loyalty Duration (81.95% importance), Customer Spend, and Visit Frequency are the strongest predictors of churn in our model.

5. Customer Segmentation & Targeting

Business Question: Which customer segments should we target with marketing campaigns?
We applied K-Means clustering (selected over Bisecting K-Means due to a superior Silhouette score of 0.2834) and identified k=6 distinct segments. Interestingly, five of the six clusters represent an "average" customer with a ~22% churn rate and ~8 visits. However, two segments stand out:

- Cluster 2 (The VIPs): 20,591 highly loyal customers averaging 8.2 loyalty years and 22.7 visits. They have our lowest churn rate (14.4%). Action: Protect this segment with loyalty tier upgrades and early-access rewards to maintain their loyalty.

- Clusters 1 & 3 (High Risk): These groups exhibit the highest churn rates (24.6% and 25.0%) and the lowest loyalty (~1.6 to 1.8 years). Action: Allocate win-back discount campaigns and re-engagement emails specifically to these segments.

6. Deployment & Reproducibility

To prepare for production, we built a reusable PCA pipeline using standardised numerical features (Quantity, Price, Total Amount, Age, Loyalty, Visits). This pipeline reduced the feature space from 6 to 4 principal components while retaining 94.94% of the variance, simplifying downstream analysis. The pipeline was saved as a .joblib artifact and successfully tested for load-and-transform reproducibility, ready for integration into RetailMart's batch-scoring environment.

Conclusion

By implementing the GBT model and targeting the specific customer segments identified in this analysis, RetailMart can transition from reactive to proactive customer retention, minimizing churn risk and maximizing long-term customer value.